CLASS 10 MATHEMATICS :: SLIP TEST – FA-1

Maximum Marks: 20 • Time: 45 Minutes

Name: _____ **Roll No:** _____ **Sec:** _____ **Marks:** _____

SECTION I: Short Answer Questions — 3 × 2M = 6 Marks

1. Find the HCF and LCM of 72 and 120 using the prime factorisation method. **[2 M]**

2. If A = {2, 3, 5, 7, 11} and B = {1, 3, 5, 7, 9}, find A ∪ B and A ∩ B. **[2 M]**

3. Find the zeroes of the quadratic polynomial $p(x) = x^2 - 2x - 8$ by factorisation. **[2 M]**

SECTION II: Long Answer Questions — 2 × 4M = 8 Marks

4. Prove that $5 - 2\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number. **[4 M]**

5. Let μ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {2, 4, 6, 8, 10}, and B = {3, 6, 9}. **[4 M]**
(i) Find A ∪ B and A ∩ B.
(ii) Find A − B and B − A.
(iii) Verify that n(A ∪ B) = n(A) + n(B) − n(A ∩ B).

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Multiple Choice Section

SECTION III: Multiple Choice Questions **6 × 1M = 6 Marks**

6. Which of the following is a terminating decimal?

(A) 7/15 (B) 13/125 (C) 11/30 (D) 17/70

7. The value of $\log_2 64$ is:

(A) 5 (B) 6 (C) 7 (D) 8

8. If n(A) = 12, n(B) = 8, and n(A ∩ B) = 3, then n(A ∪ B) is:

(A) 17 (B) 23 (C) 20 (D) 14

9. If A and B are disjoint sets, then A ∩ B is equal to:

(A) A (B) B (C) ∅ (D) μ

10. The sum and product of zeroes of a quadratic polynomial are −3 and 2. The polynomial is:

(A) $x^2 - 3x + 2$ (B) $x^2 + 3x + 2$ (C) $x^2 + 3x - 2$ (D) $x^2 - 3x - 2$

11. If α and β are zeroes of $p(x) = 2x^2 - 5x + 7$, then the value of αβ is:

(A) 5/2 (B) −5/2 (C) 7/2 (D) −7/2

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-1

Section I & Section II Solutions

SECTION I: Short Answer Solutions — 6 Marks

Solution 1 (HCF & LCM) — 2 M

$72 = 2^3 \times 3^2$

$120 = 2^3 \times 3^1 \times 5^1$

HCF $= 2^3 \times 3^1 = 8 \times 3 = 24$

LCM $= 2^3 \times 3^2 \times 5^1 = 8 \times 9 \times 5 = 360$

HCF = 24 | LCM = 360

Solution 2 (Set Operations) — 2 M

A = {2, 3, 5, 7, 11}, B = {1, 3, 5, 7, 9}

A ∪ B = {1, 2, 3, 5, 7, 9, 11}

A ∩ B = {3, 5, 7}

A ∪ B = {1, 2, 3, 5, 7, 9, 11} | A ∩ B = {3, 5, 7}

Solution 3 (Zeroes of Polynomial) — 2 M

$x^2 - 2x - 8 = 0$

$x^2 - 4x + 2x - 8 = 0$

$x(x - 4) + 2(x - 4) = 0$

$(x - 4)(x + 2) = 0$

x − 4 = 0 ➔ x = 4 | x + 2 = 0 ➔ x = −2 ➔ Zeroes are 4, −2

SECTION II: Long Answer Solutions — 8 Marks

Solution 4 (Irrationality Proof) — 4 M

Assume $5 - 2\sqrt{3}$ is rational. Then $5 - 2\sqrt{3} = a/b$ (where a, b are integers and $b \neq 0$).

$5 - (a/b) = 2\sqrt{3}$

$(5b - a)/b = 2\sqrt{3}$

$\sqrt{3} = (5b - a)/(2b)$

Since a and b are integers, $(5b - a)/(2b)$ is rational, which implies $\sqrt{3}$ is rational.

This contradicts the given fact that $\sqrt{3}$ is irrational.

Hence, $5 - 2\sqrt{3}$ is an irrational number.

Solution 5 (Sets Cardinality) — 4 M

μ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {2, 4, 6, 8, 10}, B = {3, 6, 9}

(i) A ∪ B = {2, 3, 4, 6, 8, 9, 10} | A ∩ B = {6}

(ii) A − B = {2, 4, 8, 10} | B − A = {3, 9}

(iii) n(A) = 5, n(B) = 3, n(A ∩ B) = 1, n(A ∪ B) = 7

n(A) + n(B) − n(A ∩ B) = 5 + 3 − 1 = 7

Since 7 = 7, the relation n(A ∪ B) = n(A) + n(B) − n(A ∩ B) is verified.

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-1

Section III Multiple Choice Solutions

SECTION III: Multiple Choice Solutions — **6 Marks**

Solution 6 — **1 M**

$125 = 5^3$ (form 2^n5^m)

13/125 terminates.

Answer: (B) 13/125

Solution 7 — **1 M**

$\log_2 64 = \log_2(2^6)$

$= 6 \log_2 2 = 6(1) = 6$

Answer: (B) 6

Solution 8 — **1 M**

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 12 + 8 - 3 = 17$

Answer: (A) 17

Solution 9 — **1 M**

Disjoint sets have no common elements.

$A \cap B = \varnothing$

Answer: (C) ∅

Solution 10 — **1 M**

$p(x) = x^2 - (\alpha + \beta)x + \alpha\beta$

$= x^2 - (-3)x + 2 = x^2 + 3x + 2$

Answer: (B) $x^2 + 3x + 2$

Solution 11 — **1 M**

Product of zeroes $\alpha\beta = c/a$

$c = 7$, $a = 2$ ➔ $\alpha\beta = 7/2$

Answer: (C) 7/2